Alistair Turcan

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Education

Ph.D. Computational Biology, School of Computer Science, Carnegie Mellon University — Aug 2022 – Expected 2027
Advisor: Martin Jinye Zhang

Experience

Research Intern, Phylo — Summer 2026
Mentor: Kexin Huang

Research Interests

I develop statistical and machine-learning methods for large-scale genomic data, with a focus on connecting genetic variation to disease mechanisms at cellular resolution. I then build agentic AI systems that automate key parts of this process, such as method development, code optimization, and benchmark design. Specific topics include:

Statistical genetics*:* genetic architecture of diseases and complex traits; functional interpretation of GWAS variants; linking variants to target genes and cell types

Agentic AI for biology*:* AutoResearch; biomedical agents; AI code optimization

Distinctions

1. Reviewers' Choice Abstract (Top 10%), *ASHG*, 2026, for the abstract "TusoAI-CB: automatic end-to-end optimization of computational biology methods with agentic AI"
2. Reviewers' Choice Abstract (Top 10%), *ASHG*, 2024, for the abstract "Powerful and Tagging-Aware Polygenic Enrichment Identifies Disease Associations of Individual Cells"
3. Center for Machine Learning and Health Fellowship ($150,000), *Carnegie Mellon University*, 2024–2025, for "Complex Genetic Modelling to Enable Precision Medicine for Heritable Disease"
4. Travel Award ($1,500), *Carnegie Mellon University*, 2026

Publications

(* equal contribution) († co-senior authors)

Key-author publications

1. **Alistair Turcan**, Kangcheng Hou, Kevin Z. Lin, Andreas Pfenning, Saori Sakaue, Martin Jinye Zhang. "Conditional polygenic enrichment distinguishes causal from tagging disease-critical cell populations in single-cell RNA-seq", *medRxiv*, 2026. [link] Under review at *Nature Genetics*. Related abstract awarded ASHG 2024 Reviewers' Choice (Top 10%), related research awarded $150,000 Center for Machine Learning and Health fellowship.
2. **Alistair Turcan**, Elizabeth Dorans, Kangcheng Hou, Jane Siwek, Jishnu Das, Maria Chikina, Luca Pinello, Lei Li, Alkes Price†, Kexin Huang†, Martin Jinye Zhang†. "Towards automated development of computational biology methods", *in preparation.* Related abstract awarded ASHG 2026 Reviewers' Choice (Top 10%). Currently deployed on Biomni Lab, used by 100+ researchers and several top pharma companies.
3. **Alistair Turcan***, Kangcheng Hou*, Alkes Price†, Martin Jinye Zhang†. "Recovering cell-type specific variation in RNA-seq data", *in preparation.*
4. **Alistair Turcan**, Kexin Huang, Lei Li, Martin Jinye Zhang. "TusoAI: Agentic Optimization for Scientific Methods", *ICLR*, 2026. [link]
5. Ruiqi Lyu*, **Alistair Turcan***, Bryan Wilder. "SpatialEpiBench: Benchmarking Spatial Information and Epidemic Priors in Forecasting", *NeurIPS*, 2026. [link]
6. Ruiqi Lyu*, **Alistair Turcan***, Martin Jinye Zhang†, Bryan Wilder†. "Improving constraint-based discovery with robust propagation and reliable LLM priors", *NeurIPS*, 2026. [link]
7. Ruiqi Lyu, **Alistair Turcan**, Bryan Wilder. "Explaining Concept Shift with Interpretable Feature Attribution", *ICML*, 2026. [link]
8. Ruiqi Lyu, **Alistair Turcan**, Bryan Wilder. "Population-Robust Feature Selection via Generalized Welfare Optimization", *arXiv*, 2026. [link]
9. Ruiqi Lyu, **Alistair Turcan**, Bryan Wilder. "Combining digital data streams and epidemic networks for real time outbreak detection", *arXiv*, 2025. Under revision at *PNAS*. [link]

10. **Alistair Turcan**, Jiwon Lee, Aaron Wacholder, Anne-Ruxandra Carvunis. “Integrative Detection of Genome-wide Translation using iRibo”, *STAR Protocols*, 2024. [link]
11. **Alistair Turcan**, Grant Chou, Lilu Martin, Theo Miller, Dylan Thompson, Filip Jagodzinski. “Exhaustive In-silico Simulation of Single Amino Acid Insertion and Deletion Mutations”, *IEEE BIBM*, 2022. [link]
12. Muneeba Jilani, **Alistair Turcan**, Nurit Haspel, Filip Jagodzinski. “Elucidating the Structural Impacts of Protein InDels”, *Biomolecules*, 2022. [link]
13. Muneeba Jilani, **Alistair Turcan**, Nurit Haspel, Filip Jagodzinski. “Assessing the Effects of Amino Acid Insertion and Deletion Mutations”, *IEEE BIBM*, 2021. [link]
14. **Alistair Turcan**, Anna Zivkovic, Dylan Thompson, Lorraine Wong, Lauren Johnson, Filip Jagodzinski. “CGRAP: A Web Server for Coarse-Grained Rigidity Analysis of Proteins”, *Symmetry*, 2021. [link]

Other publications

1. Shuaike Shen, Wenduo Cheng, Mingqian Ma, **Alistair Turcan**, Martin Jinye Zhang, Jian Ma. “SkillFoundry: Building Self-Evolving Agent Skill Libraries from Heterogeneous Scientific Resources”, *COLM*, 2026. [link]
2. James L. Li, Maria Zanti, Jacob Williams, Om Jahagirdar, Guochong Jia, **Alistair Turcan**, …, Haoyu Zhang. “Genetic and Cellular Architecture of Breast Cancer Risk in Multi-Ancestry Studies of 159,297 Cases and 212,102 Controls”, *JNCI*, 2025. [link]
3. Kevin Ren, Santiago Cortes-Gomez, Carlos Miguel Patiño, Ananya Joshi, Ruiqi Lyu, Jingjing Tang, **Alistair Turcan**, Khurram Yamin, Steven Wu, Bryan Wilder. “Predicting Language Models' Success at Zero-Shot Probabilistic Prediction”, *Findings of EMNLP*, 2025. [link]
4. Yixiang Qu[*], Ruiqi Lyu[*], Duan Wang, Yifan Dai, **Alistair Turcan**, Shilin Yu, Jialiu Xie, Jeffrey Roach, Catherine Butler, Pew-Thian Yap, Hongtu Zhu, Stuart Dashper, Apoena Aguiar Ribeiro, Didong Li, Kimon Divaris, Di Wu. “Deep-learning-based interpolation of longitudinal microbiome data powers biologically informative discovery”, *bioRxiv*, 2025. [link]
5. Qiang Hao, David Smith, Lu Ding, Amy Ko, Camille Ottaway, Jack Wilson, Kai Arakawa, **Alistair Turcan**, Timothy Poehlman, Tyler Greer. “Towards Understanding the Effective Design of Automated Formative Feedback for Programming Assignments”, *Computer Science Education*, 2022. [link]

Teaching Experiences

1. Graduate Teaching Assistant, Machine Learning for Scientists (02-620), Carnegie Mellon University, Spring 2024.
2. Graduate Teaching Assistant, Introduction to Bioinformatics Programming with Python, University of Pittsburgh, Fall 2023.

Professional Service

1. Conference/workshop paper reviewer for RECOMB (2025, 2026), ISMB (2024), ICLR MLGenX (2024, 2026), and ICML AI4Science (2024).
2. Journal paper reviewer for PLOS Computational Biology (2 papers, 2024–2025) and NAR Genomics and Bioinformatics (2024).

References

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